Phụ lục 1
Annex 1

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INTERNSHIP OUTLINE FOR BACHELOR/ MASTER STUDENT

Department: Department of Information and Communication Technology

Topic:

Enhanced Few-Shot Open-Set Keyword Spotting with Noise-Robust Prototype Classification and Real-Time Streaming Inference

Description:

Keyword Spotting (KWS) is the task of detecting predefined spoken keywords in audio streams — the core technology behind voice-activated systems such as "Hey Siri" and "OK Google." Traditional KWS systems require thousands of labeled samples per keyword and can only recognize a fixed vocabulary defined during training.

This project builds a KWS system that overcomes these limitations through:

- **Few-Shot Learning:** Recognize new keywords from only 3–5 audio samples without retraining, using Prototypical Networks where each keyword is represented by the average embedding (prototype) of its few samples.
- **Open-Set Recognition:** Reject unknown words using an acceptance-radius threshold around each prototype — if the L2 distance exceeds this threshold, the input is classified as "unknown."

The main contribution is a **Direct L2 distance classification** method replacing probability-based scoring, providing a more intuitive open-set rejection mechanism with a clear geometric meaning.

System pipeline:

1. Audio preprocessing: raw audio (16 kHz, 1s) → MFCC features (40 computed, 10 retained).
2. Feature encoding: DSCNN-L (276 channels) → 276-dim embeddings, trained with Triplet Loss on MSWC English.
3. Classification: Open Nearest Class Mean (openNCM) with Direct L2 distance.
4. Noise robustness: DEMAND noise augmentation (SNR = 5 dB) + optional denoising at inference.
5. Streaming inference: Silero VAD + sliding window (1s window, 0.5s stride).

Datasets: Google Speech Commands v2 (evaluation), MSWC English 763 words (training), DEMAND (noise).

Tools: Python 3.10+, PyTorch 2.0+, torchaudio, Gradio (demo), Google Colab (GPU training).

Objectives:

The tasks include:

1. Build a fully functional few-shot open-set KWS system capable of learning new keywords from 3–5 audio samples without retraining the neural network.
2. Implement and evaluate the proposed Direct L2 distance classification method against a probability-based baseline, using DET curves, AUC, and ACC@5%FAR across multiple protocols (GSC Fixed, GSC Randomized, MSWC Randomized).
3. Develop a noise robustness module with training-time noise augmentation (DEMAND dataset) and benchmark KWS accuracy at different SNR levels (0, 5, 10, 20 dB, clean).
4. Implement a real-time streaming inference module integrating Voice Activity Detection (Silero VAD) and sliding-window keyword detection with end-to-end latency under 100 ms.
5. Build an interactive Gradio web demo supporting keyword enrollment, offline detection, and simulated streaming detection.
6. Deliver complete source code, trained model checkpoints, evaluation results, and a written thesis report.

Hà Nội, 3/ 4 / 2026

Student

(Full-name and signature)

An

Phạm Hoàng An